Verifying Distributed Algorithms with Interactive Theorem Provers: A Survey of Approaches

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Formal verification is a topical field in modern Computer Science. Proving the correctness of programs is not only elegant in a theoretical sense, but is often necessary in mission-critical applications. Due to the difficulty in formalizing programs, reliability in industry is often left to automated testing. However, in specific circumstances where correctness must be ensured, formal proofs may be required.

CCS Concepts: • **Theory of computation → Logic and verification**.

Additional Key Words and Phrases: distributed systems, formal verification, proof assistants

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1 INTRODUCTION

In an increasingly interconnected world, distributed algorithms play an integral part in much of the software we use on a daily basis. Ensuring these algorithms function as intended protects against failures that may be technically, economically, or even medically, critical.

1.1 Motivation

Distributed systems are famously difficult to implement correctly. Their divided and concurrent nature can often lead to subtle bugs that are difficult to reproduce. Additionally, errors can propogate throughout a system, and partial failures can lead to unexpected negative outcomes. For this reason, it is often unrealistic to acheive total coverage with traditional testing. Formal verfication can provide a strong guarantee of correctness in these situations. Interactive theorem proving as a verification method is specificly strong for distributed systems, as it can verify correctness in all states without iteratively exploring them. Theorem proving also allows for flexibility in the paramaterization of systems, such as verifying an algorithm for any N nodes.

1.2 Contributions

For this reason, this survey will explore the verification of distributed algorithms with interactive theorem provers. It provides a taxonomy of different theorem-based verfication approaches and frameworks, reviews several common tools, and contrasts with other verification approaches.

In addition, this survey will compare the theorem proving verification approach against other common methods, such as manual proofs and model checking.

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2 METHODOLOGY

2.1 Literature Search Strategy

To identify relevant literature, this survey employed a systematic literature search strategy. Two major academic databases were selected, the ACM digital library and IEEE Xplore. These databases were chosen due to their broad coverage of research in fomral verification and distributed systems, and were accessed on April 17 2026.

To ensure the quality of the review, the PRISMA framework was used to inform the search strategy. The strategy can be summarized into two steps. First, we curated relevant keywords and constructed seach queries to compile relevant papers based on the guiding questions of the survey. This led to the following seach query:

> ("proof assistant" OR "theorem prover" OR "mechanized verification" OR "formal verification") AND ("distributed protocol" OR "distributed algorithm")

Selected papers were limited to peer-reviewed English language publications. There was no strict lower bound imposed on the publication date, although results were naturally concentrated in the last 20 years. This query yeilded 213 results from the ACM Digital Library and 49 results from IEEE Xplore, which were exported for further screening.

2.2 Inclusion and Exclusion Criteria

These results were then screened by title and abstract with standard inclusion and exclusion requirements. Included papers focused on verfication of distributed algorithms using theorem proving. Duplicate papers within and between databases were removed. Additionally, any papers focused on model checking or manual proofs were removed. This reduced the number of results to 37 for ACM and 15 for IEEE Xplore.

After this, the reference lists of the selected papers were scanned and fundamental or commonly referenced papers were added. Google scholar was used as a supplementary tool for citation chaining. Any additional papers identified this way were included only if they met the above criteria, or primarily concerned a related topic to be explained.

3 PRELIMINARY

3.1 Distributed Algorithms

Distributed computing is a fundamental field in Computer Science. It concerns the inter-communication between distinct autonomous components in a system. This patterns shows up often, usually when there is a network of multiple devices.

There are several common challenges in distributed computing that are often the targets of formal verification. For example, consensus protocols such as Paxos Another example is Byzantine Fault Tolerance.

3.2 Interactive Theorem Provers

Interactive Theorem Provers, or proof assistants, have seen a recent surge in popularity among computer scientists and mathematicians alike. Proof assistants are tools used to assist with the human development of formal proofs. These formal proofs can then be automatically checked by the machine, ensuring correctness at each stage of the process. Theorem provers are extremely useful in the formal verification of protocols, where they can be used to ensure specific properties of a sytem hold. For this reason proof assistants such as Coq, Lean, Isabelle, and Dafny have become popular for verifying distributed algorithms.

3.3 Verification

Verifying the correctness of a program requires that specific conditions about a system can be shown to hold in certain states. These conditions can be arbitrairily chosen, but usually represent ... The two most notable property categories of a system are “Safety” and “Liveness”. A safety property ensures protection against unfavourable events, whereas liveness properties make sure that the program functions as intended.

Fault tolerance is another notable aspect of ensuring a program works as intended. Byzantine fault tolerance, specifically, is a common problem in

4 APPROACHES TO VERIFYING DISTRIBUTED ALGORITHMS

4.1 Proof techniques

Inductive invariants Temporal logic based Refinement based Compositional [12] Automation

4.2 Tools and Common frameworks

There are many languages and frameworks used for the verification of distributed systems. The most commonly used tool in the literature is Rocq, formerly Coq. There are multiple frameworks implemented for Rocq that specifically focus on distributed systems, such as Verdi [11], and Diesel [9]. These ...

Another theorem prover TLAPS [2]

Iron Fleet [4]

A more contemporary tool is Lean4. There is less research in verification with lean4, as it is a younger tool. In 2025, the Veil [6] framework was introduced as a basis to verify distributed protocols specifically.

5 CASE STUDIES

5.1 Consensus

[3],

5.2 Byzantine Fault Tolerance

Another fundamental problem in distributed computing is Byzantine Fault Tolerance. This problem was originally introduced by Lamport et. al [5], and addresses partial malfunction in a distributed system. The problem

BFT [12]

5.3 Blockchain variants

Blockchain consensus [7] Blockchain Byzantine [10]

5.4 Verified software

AWS [1] Mongo [8]

Nam id fermentum dui. Suspendisse sagittis tortor a nulla , in pulvinar ex pretium. Sed interdum orci quis metus euismod, et sagittis enim maximus. Vestibulum gravida massa ut suscipit

6 OTHER VERIFICATION STRATEGY

In contrast to using an interactive theorem prover, there are several other methods one might use to formally verify a distributed system. Each of these other verification approaches has its own advantages and disadvantages in relation to theorem proving. The two other predominant methods of formally verifying distributed systems are using model checking and manual proofs.

Many solutions to the previously mentioned problems are based on manual proofs. This makes sense, as manual proofs are much more flexibile than the other two methods.

Model checking is another approach for verifying the correctness of a distributed system. Model checking expands the state space, exhaustively exploring every possible state and ensuring that the correctness properties hold in each one. Examples of

Theorem proving also has its relative merits. One of the strengths of using an interactive theorem prover is that the state space expansion problem is not an issue. Invariant based theorems will necessarily hold for any transition between states, so there is no need to check them if a system has been verified in this way.

7 CHALLENGES AND FUTURE DIRECTIONS

One of the common criticisims of theorem based formal verification is that it is too difficult to apply practically. There are differing perspectives on this .

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8 CONCLUSION

This paper aims to provide an overview of different techniques and problems in the formal verfication of distributed systems using theorem proving.

Interactive theorem provers provide the strongest guarantee of correctness, but are still expensive when compared to other methods.

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